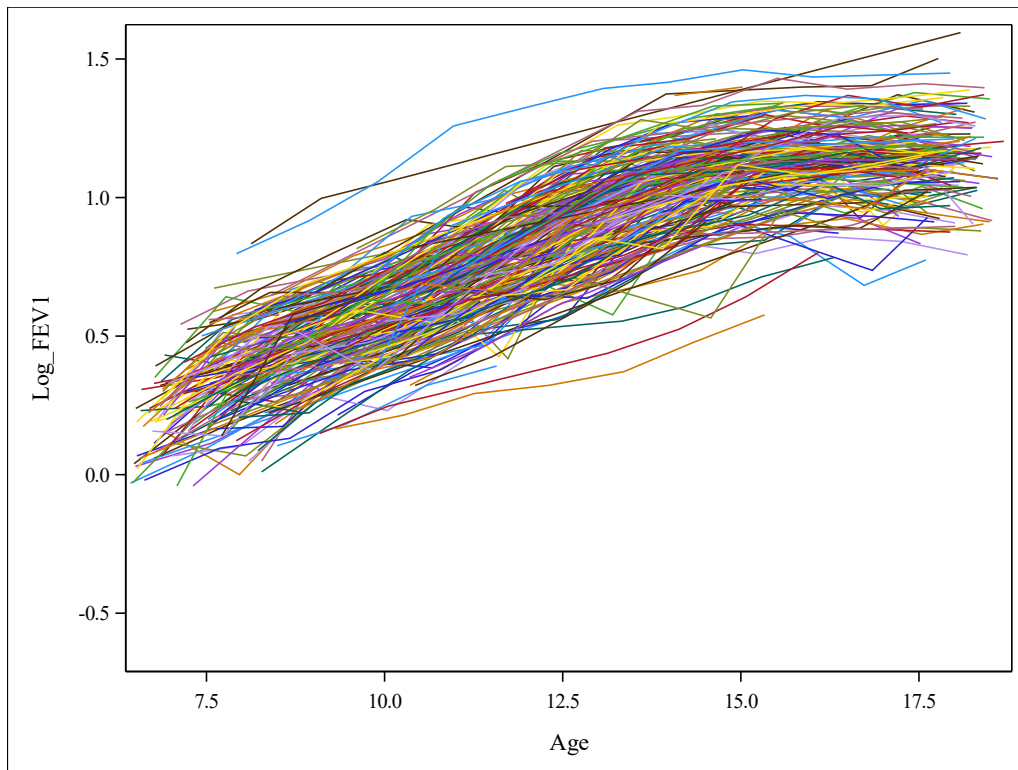


*The Six Cities Study of Air Pollution and Health was a longitudinal study designed to characterize lung growth as measured by changes in pulmonary function in children and adolescents, and the factors that influence lung function growth.

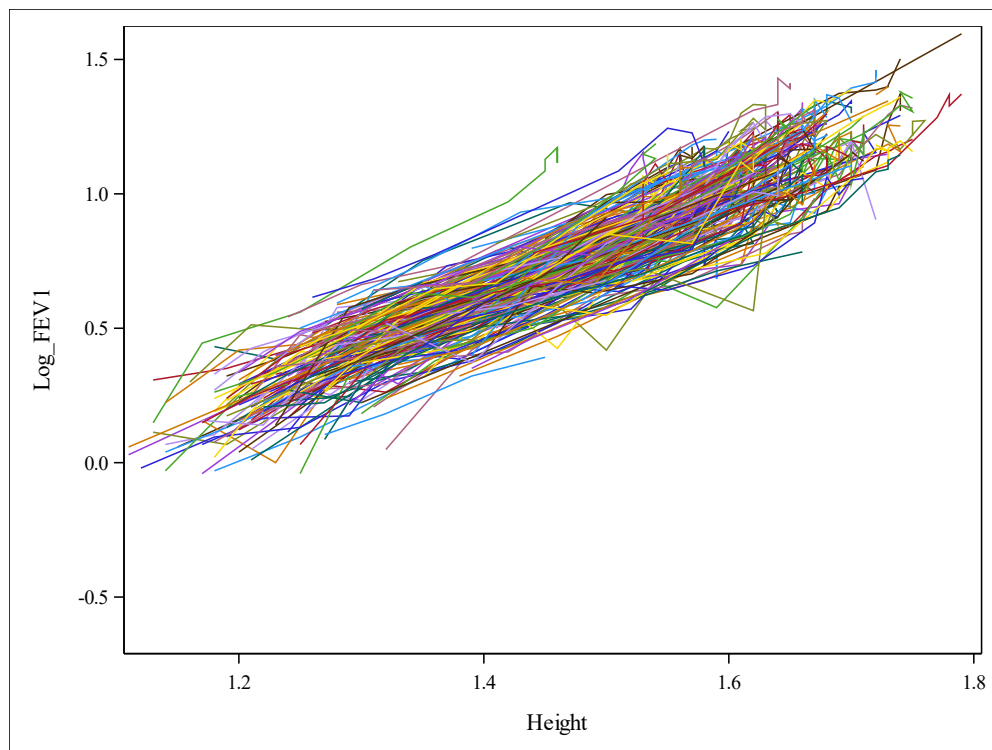
A cohort of 13,379 children born on or after 1967 was enrolled in six communities across the U.S.: Watertown (Massachusetts), Kingston and Harriman (Tennessee), a section of St. Louis (Missouri), Steubenville (Ohio), Portage (Wisconsin), and Topeka (Kansas). Most children were enrolled in the first or second grade (between the ages of six and seven) and measurements of study participants were obtained annually until graduation from high school or loss to follow-up. At each annual examination, spirometry, the measurement of pulmonary function, was performed and a respiratory health questionnaire was completed by a parent or guardian.

```
data air_pol;
input  ID Height Age INI_Height  INI_Age  Log_FEV1;
L_INI_Height = log(INI_Height);
L_Age = log(Age);
L_INI_Age = log(INI_Age);
Age_fl = floor(Age);
Height_C = Height - 1.5;
datalines;
      1      1.20      9.3415      1.20      9.3415      0.21511
      1      1.28     10.3929      1.20      9.3415      0.37156
....
    300      1.62     15.9398      1.44     11.9617      1.08181
    300      1.62     17.0075      1.44     11.9617      1.12817
    300      1.63     17.8645      1.44     11.9617      1.16938
;
run;

Proc SGplot data = air_pol;
series x=Age y=Log_FEV1 / group =ID LineAttrs= (pattern=1 );
run;
```



```
Proc SGplot data = air_pol;
series x=Height y=Log_FEV1 / group =ID LineAttrs= (pattern=1 );
run;
```



```
proc mixed data = air_pol covtest;
class ID;
model Log_FEV1 = Height L_Age / solution;
random intercept L_Age /type=UN subject=ID g gcorr v vcorr;
run;
```

Convergence criteria met.

Estimated G Matrix

Row	Effect	ID	Col1	Col2
1	Intercept	1	0.04838	-0.01823
2	L_Age	1	-0.01823	0.008342

Estimated G Correlation Matrix

Row	Effect	ID	Col1	Col2
1	Intercept	1	1.0000	-0.9073
2	L_Age	1	-0.9073	1.0000

Estimated V Matrix for ID 1

Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	0.01203	0.008614	0.008654	0.008689	0.008720	0.008779	0.008802
2	0.008614	0.01221	0.008880	0.008989	0.009086	0.009272	0.009345
3	0.008654	0.008880	0.01255	0.009263	0.009419	0.009720	0.009840
4	0.008689	0.008989	0.009263	0.01296	0.009709	0.01011	0.01027
5	0.008720	0.009086	0.009419	0.009709	0.01343	0.01045	0.01065
6	0.008779	0.009272	0.009720	0.01011	0.01045	0.01457	0.01137

Estimated V Matrix for ID 1							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
7	0.008802	0.009345	0.009840	0.01027	0.01065	0.01137	0.01512
Estimated V Correlation Matrix for ID 1							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.7106	0.7044	0.6958	0.6861	0.6629	0.6525
2	0.7106	1.0000	0.7173	0.7144	0.7095	0.6949	0.6876
3	0.7044	0.7173	1.0000	0.7263	0.7258	0.7189	0.7144
4	0.6958	0.7144	0.7263	1.0000	0.7360	0.7356	0.7335
5	0.6861	0.7095	0.7258	0.7360	1.0000	0.7473	0.7472
6	0.6629	0.6949	0.7189	0.7356	0.7473	1.0000	0.7661
7	0.6525	0.6876	0.7144	0.7335	0.7472	0.7661	1.0000
Covariance Parameter Estimates							
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr Z		
UN(1,1)	ID	0.04838	0.008880	5.45	<.0001		
UN(2,1)	ID	-0.01823	0.003569	-5.11	<.0001		
UN(2,2)	ID	0.008342	0.001505	5.54	<.0001		
Residual		0.003462	0.000127	27.19	<.0001		
Fit Statistics							
-2 Res Log Likelihood		-4638.6					

Fit Statistics	
AIC (Smaller is Better)	-4630.6
AICC (Smaller is Better)	-4630.5
BIC (Smaller is Better)	-4615.8

```
proc mixed data = air_pol covtest;
class ID;
model Log_FEV1 = Height L_Age / solution;
random intercept Height/type=UN subject=ID g gcorr v vcorr;
run;
```

Fit Statistics	
-2 Res Log Likelihood	-4643.7
AIC (Smaller is Better)	-4635.7
AICC (Smaller is Better)	-4635.6
BIC (Smaller is Better)	-4620.9

```
proc mixed data = air_pol covtest;
class ID;
model Log_FEV1 = Height L_Age / solution;
random intercept Height L_Age/type=UN subject=ID g gcorr v
vcorr;
run;
```

Convergence criteria met.

Estimated G Matrix					
Row	Effect	ID	Col1	Col2	Col3
1	Intercept	1	0.06208	-0.02683	-0.00829

Estimated G Matrix					
Row	Effect	ID	Col1	Col2	Col3
2	Height	1	-0.02683	0.01378	0.004489
3	L_Age	1	-0.00829	0.004489	0.000754

Estimated G Correlation Matrix					
Row	Effect	ID	Col1	Col2	Col3
1	Intercept	1	1.0000	-0.9173	-1.0000
2	Height	1	-0.9173	1.0000	1.0000
3	L_Age	1	-1.0000	1.0000	1.0000

Estimated V Matrix for ID 1								
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7	
1	0.01175	0.008178	0.008047	0.007921	0.007815	0.007636	0.007562	
2	0.008178	0.01163	0.008191	0.008245	0.008266	0.008182	0.008164	
3	0.008047	0.008191	0.01169	0.008447	0.008558	0.008538	0.008560	
4	0.007921	0.008245	0.008447	0.01224	0.009054	0.009134	0.009214	
5	0.007815	0.008266	0.008558	0.009054	0.01282	0.009540	0.009662	
6	0.007636	0.008182	0.008538	0.009134	0.009540	0.01316	0.009887	
7	0.007562	0.008164	0.008560	0.009214	0.009662	0.009887	0.01348	

Estimated V Correlation Matrix for ID 1							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.6996	0.6864	0.6605	0.6367	0.6140	0.6008

Estimated V Correlation Matrix for ID 1							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
2	0.6996	1.0000	0.7026	0.6912	0.6772	0.6615	0.6521
3	0.6864	0.7026	1.0000	0.7062	0.6991	0.6883	0.6818
4	0.6605	0.6912	0.7062	1.0000	0.7230	0.7198	0.7174
5	0.6367	0.6772	0.6991	0.7230	1.0000	0.7346	0.7351
6	0.6140	0.6615	0.6883	0.7198	0.7346	1.0000	0.7423
7	0.6008	0.6521	0.6818	0.7174	0.7351	0.7423	1.0000

Estimated G matrix is not positive definite.

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr Z
UN(1,1)	ID	0.06208	0.01242	5.00	<.0001
UN(2,1)	ID	-0.02683	0.01507	-1.78	0.0750
UN(2,2)	ID	0.01378	0.02922	0.47	0.3186
UN(3,1)	ID	-0.00829	0.006618	-1.25	0.2105
UN(3,2)	ID	0.004489	0.01414	0.32	0.7509
UN(3,3)	ID	0.000754	0.007514	0.10	0.4600
Residual		0.003425	0.000131	26.24	<.0001

Fit Statistics	
-2 Res Log Likelihood	-4645.5
AIC (Smaller is Better)	-4631.5

Fit Statistics	
AICC (Smaller is Better)	-4631.4
BIC (Smaller is Better)	-4605.6

```
proc mixed data = air_pol method=ML;
class ID;
model Log_FEV1 = Height L_Age / solution;
random intercept Height/type=UN subject=ID g gcorr v vcorr;
run;
```

Fit Statistics	
-2 Log Likelihood	-4664.6
AIC (Smaller is Better)	-4650.6
AICC (Smaller is Better)	-4650.5
BIC (Smaller is Better)	-4624.7

```
proc mixed data = air_pol method=ML;;
class ID;
model Log_FEV1 = Height L_Age L_Age*L_Age/ solution;
random intercept Height_C/type=UN subject=ID g gcorr v vcorr;
run;
```

Fit Statistics	
-2 Log Likelihood	-4689.1
AIC (Smaller is Better)	-4673.1
AICC (Smaller is Better)	-4673.0
BIC (Smaller is Better)	-4643.5


```
proc mixed data = air_pol method=ML;
class ID Age_fl;
model Log_FEV1 = Height Age_fl / solution;
random intercept Height/type=UN subject=ID g gcorr v vcorr;
run;
```

Fit Statistics	
-2 Log Likelihood	-4801.9
AIC (Smaller is Better)	-4765.9
AICC (Smaller is Better)	-4765.5
BIC (Smaller is Better)	-4699.2

```
proc mixed data = air_pol method=ML;
class ID Age_fl;
model Log_FEV1 = Height_C Height_C*Height_C Age_fl/ solution;
random intercept Height_C/type=UN subject=ID g gcorr v vcorr;
run;
```

Fit Statistics	
-2 Log Likelihood	-4803.2
AIC (Smaller is Better)	-4765.2
AICC (Smaller is Better)	-4764.8
BIC (Smaller is Better)	-4694.9

```
proc mixed data = air_pol;
class ID Age_fl;
model Log_FEV1 = Height Age_fl/ solution outp=air_pol_age_pred;
random intercept Height/type=UN subject=ID g gcorr v vcorr;
run;
```

The Mixed Procedure

Model Information	
Data Set	WORK.AIR_POL
Dependent Variable	Log_FEV1
Covariance Structure	Unstructured
Subject Effect	ID
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Containment

Class Level Information		
Class	Levels	Values
ID	299	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300
Age_fl	13	6 7 8 9 10 11 12 13 14 15 16 17 18

Dimensions	
Covariance Parameters	4
Columns in X	15
Columns in Z per Subject	2

Dimensions	
Subjects	299
Max Obs per Subject	12

Number of Observations	
Number of Observations Read	1993
Number of Observations Used	1993
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	-2925.36557288	
1	2	-4687.50716296	0.00004161
2	1	-4687.68868165	0.00000024
3	1	-4687.68970168	0.00000000

Convergence criteria met.

Estimated G Matrix				
Row	Effect	ID	Col1	Col2
1	Intercept	1	0.07629	-0.05131
2	Height	1	-0.05131	0.03869

Estimated G Correlation Matrix				
Row	Effect	ID	Col1	Col2
1	Intercept	1	1.0000	-0.9445
2	Height	1	-0.9445	1.0000

Estimated V Matrix for ID 1							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	0.01200	0.008465	0.008221	0.007781	0.007488	0.007390	0.007292
2	0.008465	0.01147	0.008232	0.008071	0.007964	0.007928	0.007892
3	0.008221	0.008232	0.01138	0.008252	0.008261	0.008264	0.008267
4	0.007781	0.008071	0.008252	0.01172	0.008796	0.008868	0.008941
5	0.007488	0.007964	0.008261	0.008796	0.01230	0.009272	0.009391
6	0.007390	0.007928	0.008264	0.008868	0.009272	0.01255	0.009541
7	0.007292	0.007892	0.008267	0.008941	0.009391	0.009541	0.01284

Estimated V Correlation Matrix for ID 1							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.7216	0.7033	0.6560	0.6163	0.6021	0.5875
2	0.7216	1.0000	0.7205	0.6961	0.6706	0.6608	0.6505
3	0.7033	0.7205	1.0000	0.7143	0.6981	0.6913	0.6838
4	0.6560	0.6961	0.7143	1.0000	0.7325	0.7311	0.7288
5	0.6163	0.6706	0.6981	0.7325	1.0000	0.7463	0.7474
6	0.6021	0.6608	0.6913	0.7311	0.7463	1.0000	0.7516
7	0.5875	0.6505	0.6838	0.7288	0.7474	0.7516	1.0000

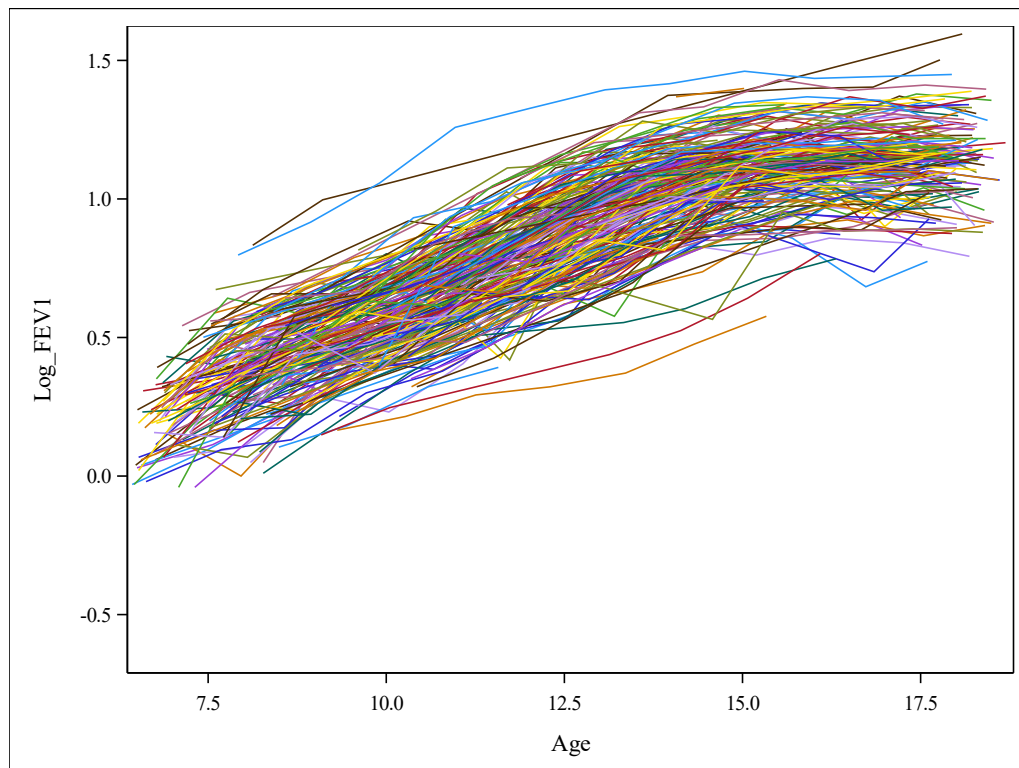
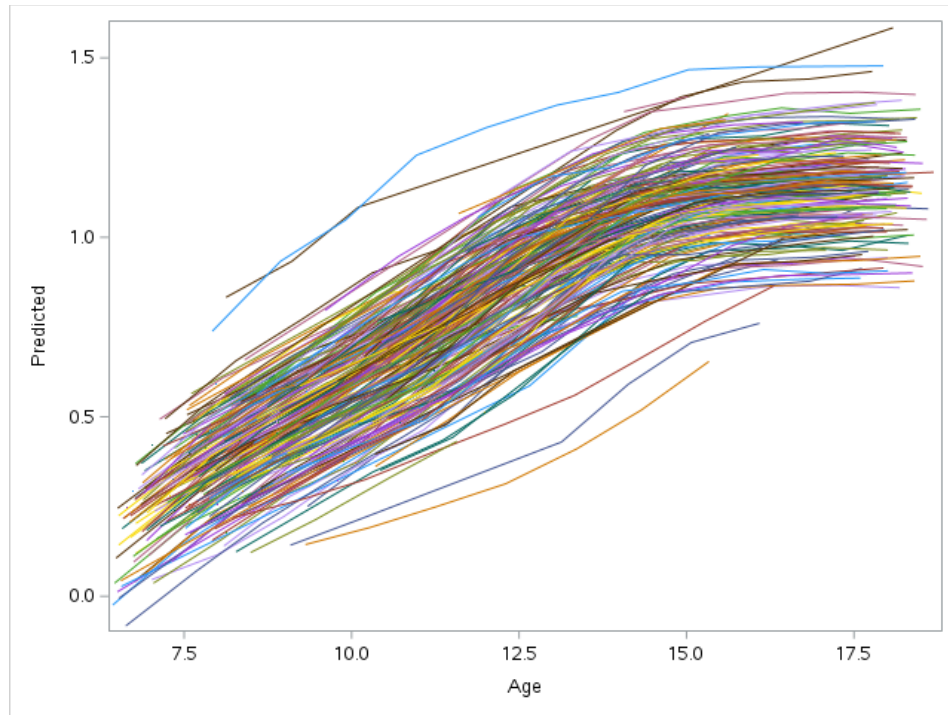
Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1,1)	ID	0.07629
UN(2,1)	ID	-0.05131

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(2,2)	ID	0.03869
Residual		0.003146
Fit Statistics		
-2 Res Log Likelihood		-4687.7
AIC (Smaller is Better)		-4679.7
AICC (Smaller is Better)		-4679.7
BIC (Smaller is Better)		-4664.9
Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
3	1762.32	<.0001

Solution for Fixed Effects						
Effect	Age_fl	Estimate	Standard Error	DF	t Value	Pr > t
Intercept		-1.6180	0.08785	298	-18.42	<.0001
Height		1.6769	0.05377	251	31.19	<.0001
Age_fl	6	-0.2116	0.02651	1430	-7.98	<.0001
Age_fl	7	-0.1556	0.02267	1430	-6.86	<.0001
Age_fl	8	-0.1356	0.01977	1430	-6.86	<.0001
Age_fl	9	-0.1258	0.01706	1430	-7.37	<.0001
Age_fl	10	-0.1200	0.01440	1430	-8.33	<.0001
Age_fl	11	-0.1179	0.01186	1430	-9.94	<.0001

Solution for Fixed Effects						
Effect	Age_fl	Estimate	Standard Error	DF	t Value	Pr > t
Age_fl	12	-0.09507	0.009837	1430	-9.66	<.0001
Age_fl	13	-0.06758	0.008884	1430	-7.61	<.0001
Age_fl	14	-0.02852	0.008516	1430	-3.35	0.0008
Age_fl	15	-0.00407	0.008368	1430	-0.49	0.6269
Age_fl	16	0.003642	0.008394	1430	0.43	0.6645
Age_fl	17	0.006674	0.008410	1430	0.79	0.4276
Age_fl	18	0	.	.	.	.
Type 3 Tests of Fixed Effects						
Effect	Num DF	Den DF	F Value	Pr > F		
Height	1	251	972.65	<.0001		
Age_fl	12	1430	30.47	<.0001		

```
Proc SGplot data = air_pol_age_pred;
series x=Age y=pred / group =ID LineAttrs= (pattern=1 );
run;
```



```
Proc SGplot data = air_pol_age_pred;  
series x=Height y=pred / group =ID LineAttrs= (pattern=1 );  
run;
```

